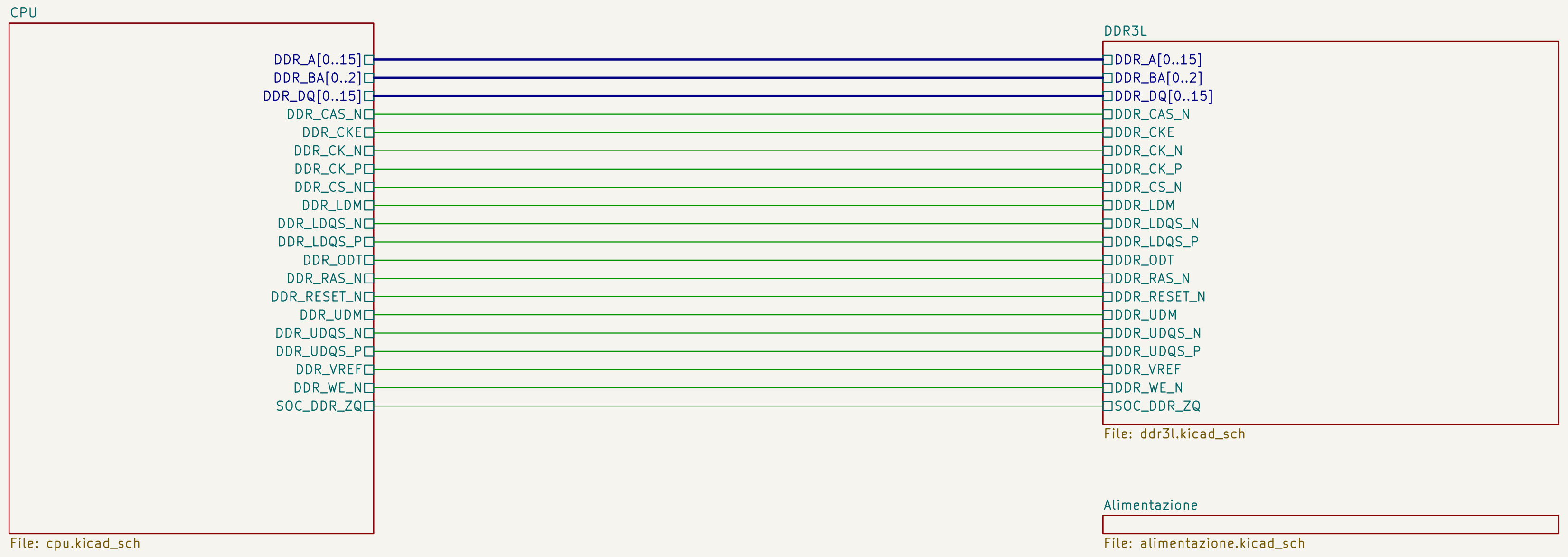
Alimentazioni e masse (3V3, 5V_SYS, GND_D, V1P35, V1P4) sono etichette globali nei fogli: non sono disegnate qui.



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U1 NXP MCIMX6Y2CVM08AB (i.MX 6ULL, MAPBGA−289 14x14 mm passo 0.8). Scheda di prova per lo sbroglio: collegati solo DDR3L, alimentazioni, clock, reset, boot e console UART1.

Boot: BOOT_MODE[1:0]=01 (serial downloader, la scheda non ha memorie di boot): il ROM resta in attesa su UART1/USB. R105/R106 DNP per BOOT_MODE=10 (boot interno).

POR_B: RC 10k/1uF dal 3.3 V (circa 10 ms). JTAG non usato: JTAG_MOD a GND con 4.7k (tabella 3).

VDD_ARM_CAP / VDD_SOC_CAP: uscite LDO interni, solo condensatori

VDD_HIGH_CAP (a NVCC_DRAM_2P5), NVCC_PLL, VDD_SNVS_CAP, VDD_USB_CAP

VDD_SOC_IN 1.4 V

VDD_HIGH_IN e NVCC 3.3 V: un 100 nF per dominio (CSI, ENET, GPIO, LCD, NAND, SD1, UART)

NVCC_DRAM 1.35 V (lato SoC)

VDD_SNVS_IN

Carico 24 MHz (CL da verificare sul quarzo YXC)

Carico RTC: circa 2xCL meno parassite (datasheet, tabella 3)

Boot: BOOT_MODE=01 serial downloader; le due DNP danno BOOT_MODE=10; JTAG_MOD a GND con 4.7k

POR_B: RC 10k/1uF

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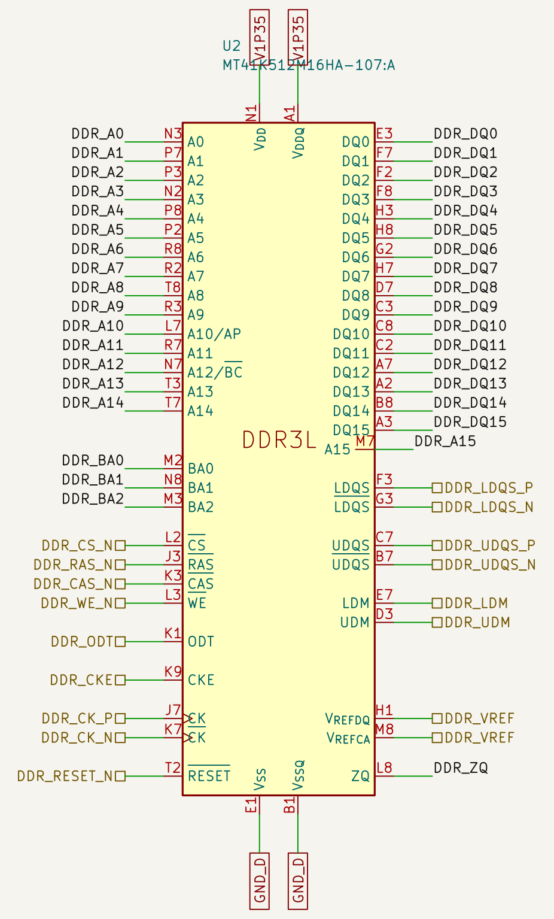
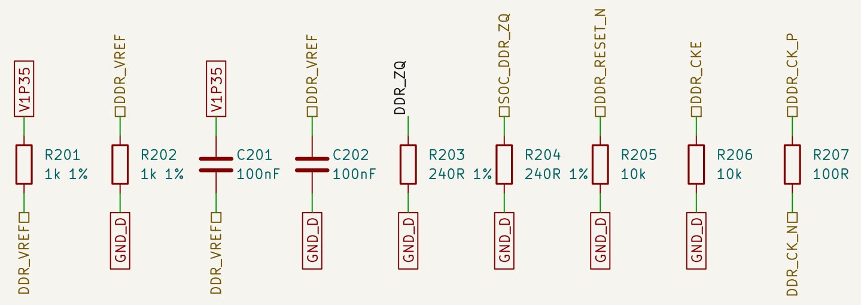
U2 Micron MT41K512M16HA-107:A, 8 Gb x16 = 1 GB, FBGA-96 9x14 passo 0.8, a 1.35 V (equivalente: Alliance AS4C512M16D3LC, stessa ball map). Un solo chip punto-punto: niente VTT. Simbolo da MT41K256M16HA di KiCad con A15 su M7.

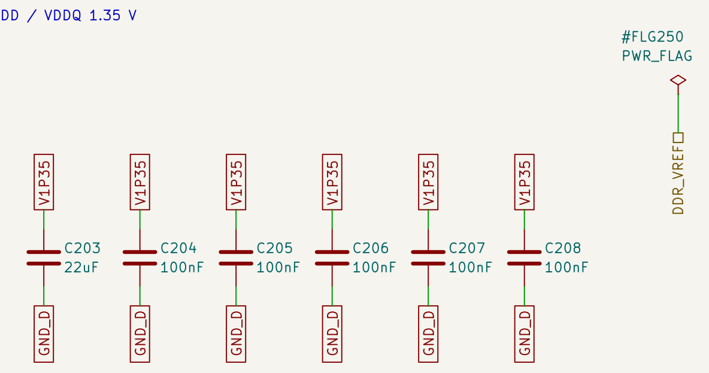
VREF a partitore 1k/1k con 100 nF su ogni resistenza, unica per DRAM_VREF, VREFCA, VREFDQ. ZQ 240R 1% su SoC e DRAM. RESET# e CKE con 10k a GND. CK con 100R differenziale vicino alla DRAM.



VREF, ZQ, pull-down



VDD / VDDQ 1.35 V



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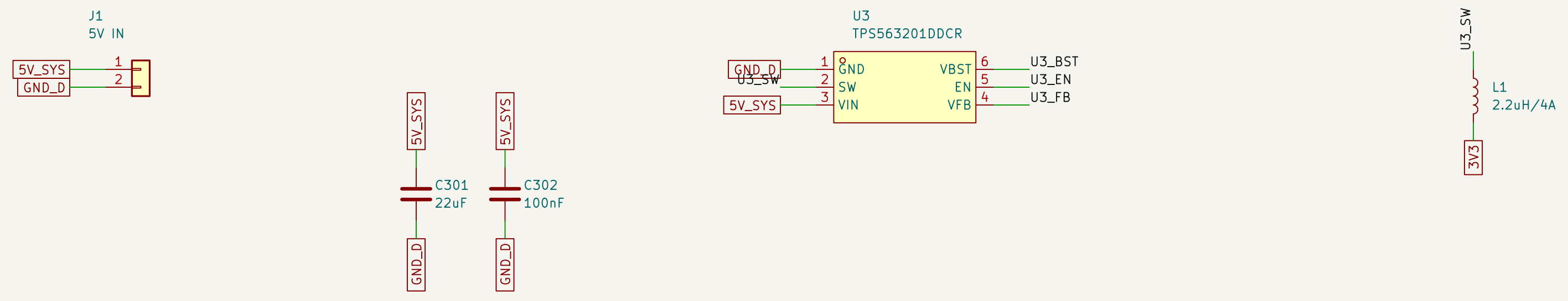
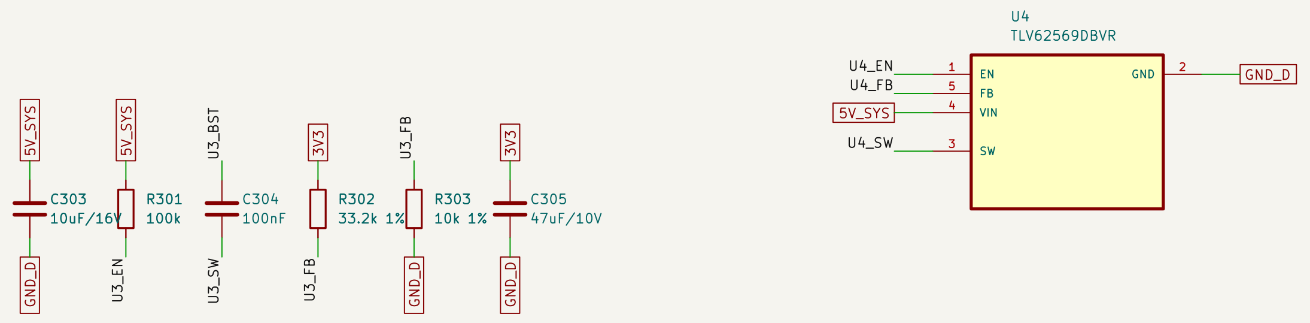
Ingresso 5 V su J1 (header 2 pin) -> buck 3.3 V (U3 TPS563201), 1.35 V DDR (U4 TLV62569), 1.4 V VDD_SOC_IN (U5 TLV62569). Stessi valori della scheda SchedaIMX6ULL-SD.

Sequenza 3.3 -> 1.35 -> 1.4 V a RC sugli EN dei TLV62569 (soglia EN max 1.2 V): 100k/470nF circa 21 ms, 100k/1uF circa 45 ms. Passivi 0603 minimo; 47 uF in 1206, 22 uF e 10 uF in 0805.

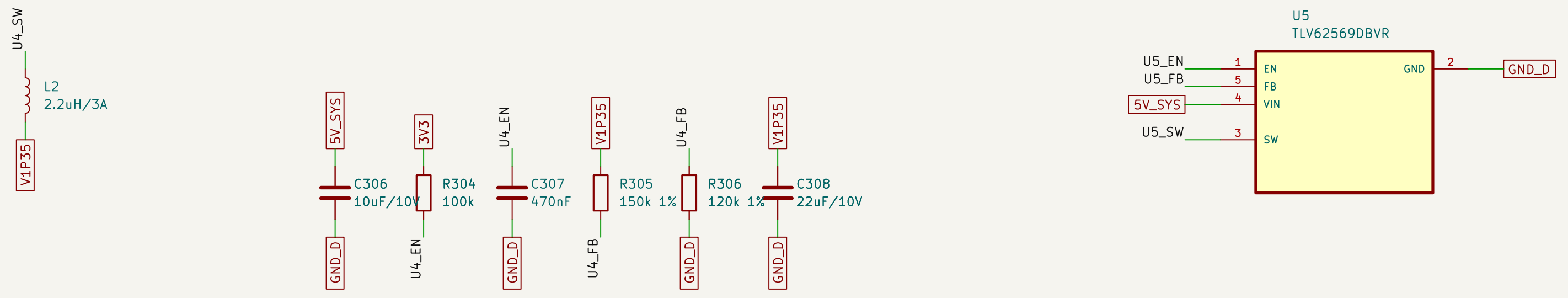
Ingresso 5 V



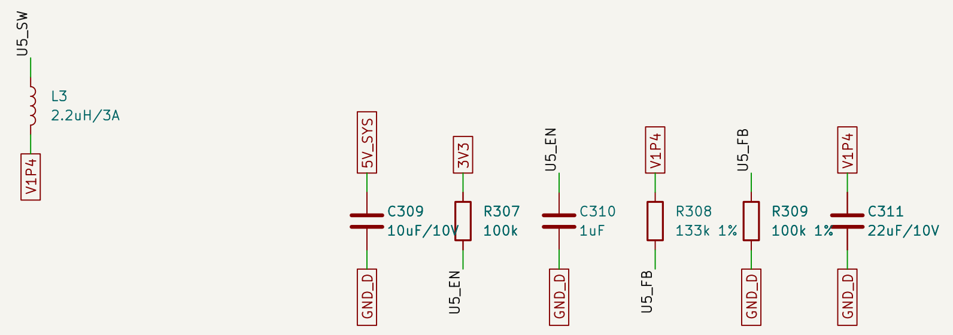
U3 3.3 V / 3 A: FB 33.2k/10k = 3.32 V, primo della sequenza

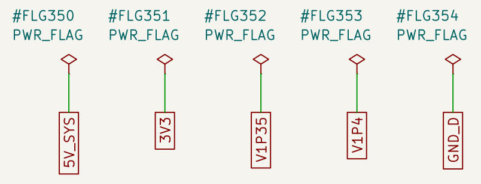


U4 1.35 V DDR: FB 150k/120k (0.6 V x 2.25), EN 100k/470nF da 3V3



U5 1.4 V VDD_SOC_IN: FB 133k/100k = 1.398 V, EN 100k/1uF da 3V3





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